Wave Motion

1. Definition of Waves

- Waves are disturbances that propagate through a medium or space, carrying energy without causing any permanent displacement of the medium itself.

- They can be characterized by their amplitude, wavelength, frequency, and speed.

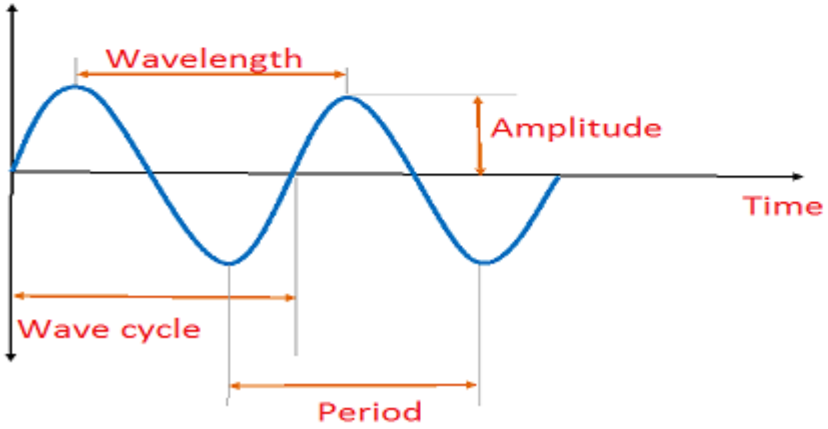
Types of Waves

- Mechanical Waves: Waves that require a medium for propagation, such as water waves and sound waves.

- Electromagnetic Waves: Waves that can propagate through a vacuum, such as light waves and radio waves.

Characteristics of Waves

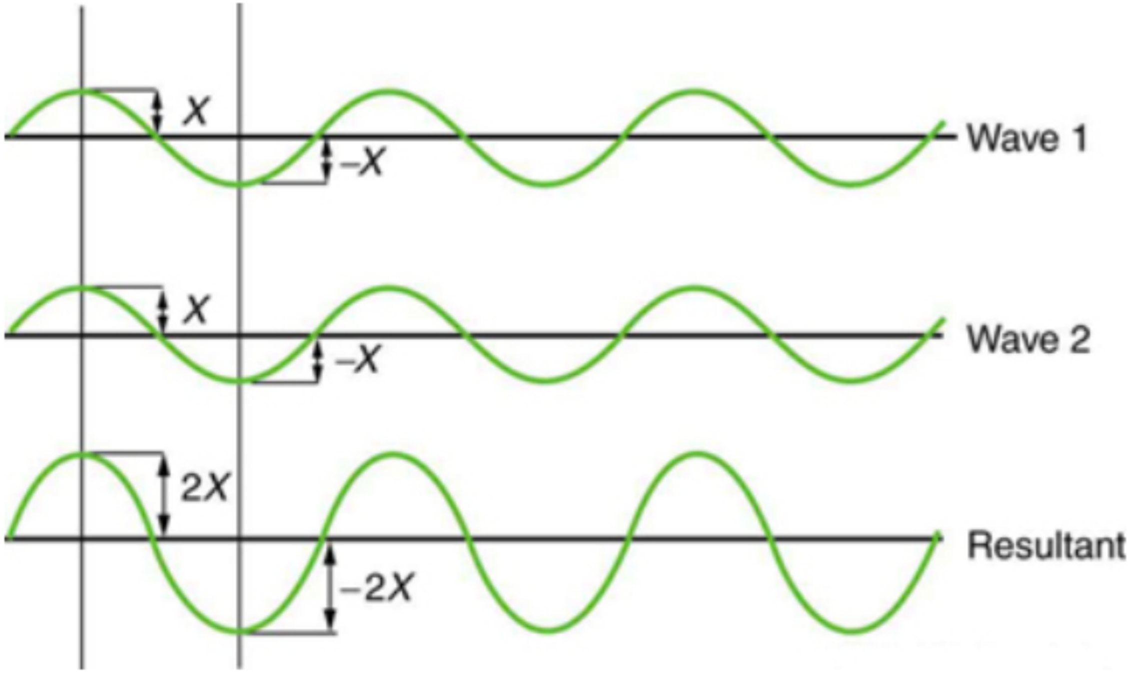
- Amplitude: The maximum displacement of a wave from its equilibrium position.
- Wavelength: The distance between two consecutive points in a wave that are in phase.
- Frequency: The number of complete oscillations or cycles of a wave per unit time.
- Period: The time taken for one complete cycle of a wave.
- Speed: The rate at which a wave travels through a medium.


Wavelength
Amplitude
Time
Wave cycle
Period

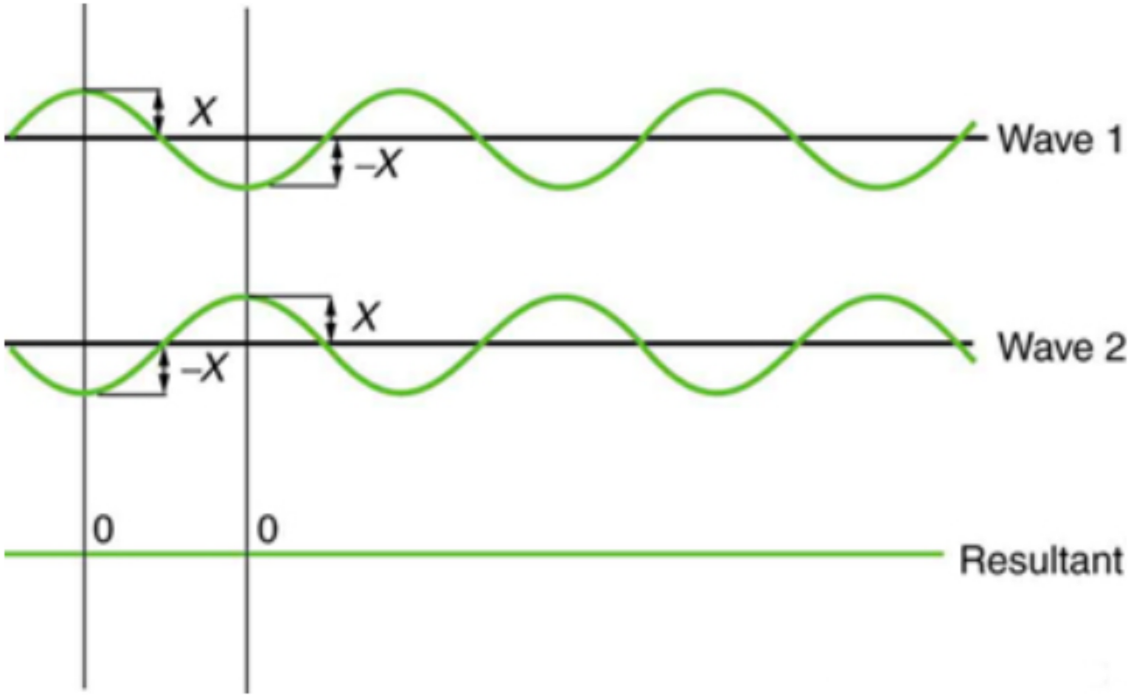
Wave Interference

- Constructive Interference: When two waves meet to produce a larger amplitude.
- Destructive Interference: When two waves meet to produce a smaller or zero amplitude.

Constructive interference



Destructive interference



Many examples in physics consist of two sinusoidal waves that are identical in amplitude, wave number, and angular frequency, but differ by a phase shift:

$$y_1(x,t) = A\sin(kx - \omega t + \varphi),$$
$$y_2(x,t) = A\sin(kx - \omega t).$$

When these two waves exist in the same medium, the resultant wave resulting from the superposition of the two individual waves is the sum of the two individual waves:

$$y_R(x,t) = y_1(x,t) + y_2(x,t) = A\sin(kx - \omega t + \varphi) + A\sin(kx - \omega t).$$

$$\sin u + \sin v = 2\sin(\tfrac{u+v}{2})\cos(\tfrac{u-v}{2}),$$

The resultant wave can be better understood by using the trigonometric identity:

$$\begin{aligned} y_R(x,t) &= y_1(x,t) + y_2(x,t) = A\sin(kx - \omega t + \varphi) + A\sin(kx - \omega t) \\ &= 2A\sin(\tfrac{(kx-\omega t+\varphi)+(kx-\omega t)}{2})\cos(\tfrac{(kx-\omega t+\varphi)-(kx-\omega t)}{2}) \\ &= 2A\sin(kx - \omega t + \tfrac{\varphi}{2})\cos(\tfrac{\varphi}{2}). \end{aligned}$$

This equation is usually written as

$$y_R(x,t) = [2A\cos(\frac{\varphi}{2})]\sin(kx - \omega t + \frac{\varphi}{2}).$$

Wave Reflection

- The bouncing back of a wave after striking a barrier or boundary.

Wave Refraction

- The bending of a wave as it passes from one medium to another.

Wave Diffraction

- The bending of waves around obstacles or through narrow openings.

Wave reflection

Wave reflection is a phenomenon in which a wave encounters an interface or boundary separating two different mediums or regions and bounces back into the medium from which it originated. When a wave encounters the boundary, a portion of the wave's energy is reflected back while the remaining energy may be transmitted through the boundary or absorbed by it. The reflection of waves follows the law of reflection, which states that the angle of incidence is equal to the angle of reflection, and it occurs for various types of waves, including water waves, sound waves, and light waves. Understanding wave reflection is essential in fields such as optics, acoustics, and seismology and has practical applications in the design of architectural spaces, the development of sonar and radar systems, and the construction of optical devices.

Fixed End Reflection

Fixed end reflection refers to the reflection of waves that occurs when a wave encounters a boundary or an interface where the end is fixed and does not allow any displacement. In the context of waves, this phenomenon is often observed in situations where the medium is constrained at one end, preventing any motion or displacement. When a wave reaches the fixed end, it is reflected back with a phase change, resulting in a pattern of interference that depends on the wavelength and the length of the medium.

Free End Reflection

Free end reflection refers to the reflection of waves that occurs when a wave encounters a boundary or an interface where the end is free to move or undergo displacement. In the context of waves, this phenomenon is observed in situations where the medium is free to move at one end, allowing the wave to be reflected without inversion or phase change. When a wave reaches the free end, it is reflected back without any change in phase, resulting in an interference pattern that depends on the wavelength and the length of the medium.

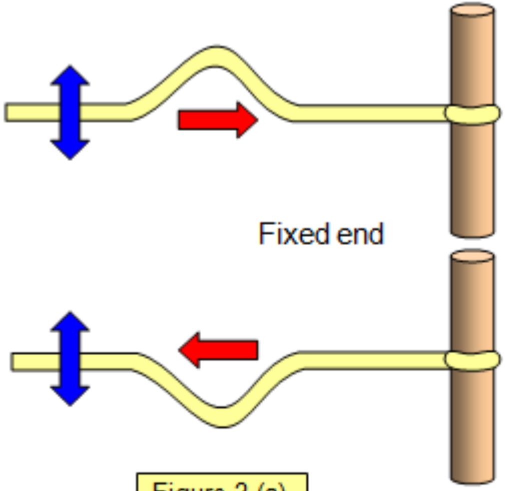


Figure 2 (a)

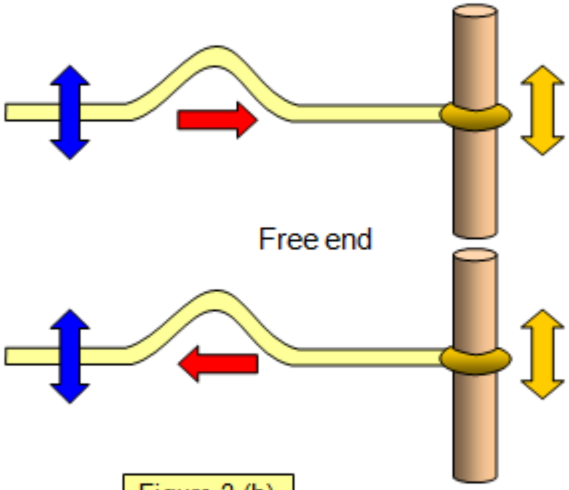


Figure 2 (b)

Wave refraction

Wave refraction is a phenomenon that occurs when a wave encounters a change in the medium it is traveling through, leading to a change in its direction as it moves from one medium to another. This change in direction is a result of the variation in the wave's speed as it transitions between mediums with different properties, such as density or elasticity.

Key characteristics of wave refraction

Change in Wave Speed: When a wave transitions from one medium to another, its speed may change, leading to a bending of the wavefront.

Change in Wave Direction: The change in speed causes the wave to change direction, typically bending either toward or away from the normal depending on whether the wave is moving into a slower or faster medium.

Snell's Law: Snell's law describes the relationship between the angles of incidence and refraction, relating the refractive indices of the two media to the angles at which the wave enters and exits the media.

Wave diffraction

Wave diffraction is a phenomenon that occurs when waves encounter an obstacle or opening that is comparable in size to their wavelength. When this happens, the waves spread out and bend around the edges of the obstacle or diffract through the opening, creating a pattern of spreading beyond the geometrical shadow of the obstacle or opening.